

Palaeogenomic inference of biodiversity dynamics across Quaternary timescales

Amanda Lindahl^{1,2} , Laura S. Epp³, Sanne Boessenkool⁴, Mikkel Winther Pedersen⁵, Selina Brace⁶, Peter D. Heintzman^{1,7}, Love Dalén^{1,2,8}  & David Díez del Molino^{1,2,8} 

Abstract

Biodiversity is essential for the resilience and stability of life, yet it is highly dynamic and has continuously evolved throughout Earth's history. The biodiversity concept encompasses three hierarchical levels of equal importance to fundamental ecological processes: diversity at the ecosystem, species and genetic levels. The current biodiversity crisis calls for an urgent need to understand the causes and consequences of widespread diversity losses at all three levels. Breakthroughs in palaeogenomics have increased the ecological and temporal scales on which we can use genomic information to study past biodiversity, reaching as far back as the Early Pleistocene. In this Review, we explore the possibilities and limitations of using palaeogenomics for studying all aspects of biodiversity. We explore how incorporating palaeogenomics into biodiversity research can provide clues about ecosystem composition, trophic interactions, species distributions, adaptation, evolution and extinction through time, in response to natural processes and as a consequence of human impact. We report how palaeogenomics can be applied to address a wide range of topics across all three hierarchical levels of biodiversity, and we show how advances within the field are making palaeogenomics an invaluable tool for understanding past and present declines in biodiversity, and in helping to predict future losses.

Sections

Introduction

Biodiversity through time

Palaeogenomics in biodiversity research

Ecosystem dynamics

Species diversity

Genetic variation

Summary and future directions

¹Centre for Palaeogenetics, Stockholm, Sweden. ²Department of Zoology, Stockholm University, Stockholm, Sweden. ³Department of Biology, University of Konstanz, Konstanz, Germany. ⁴Centre for Ecological and Evolutionary Synthesis, Department of Biosciences, University of Oslo, Oslo, Norway. ⁵Centre for Ancient Environmental Genomics, Globe Institute, University of Copenhagen, Copenhagen, Denmark. ⁶Natural History Museum, London, UK. ⁷Department of Geological Sciences, Stockholm University, Stockholm, Sweden. ⁸Department of Bioinformatics and Genetics, Swedish Museum of Natural History, Stockholm, Sweden.

 e-mail: amanda.lindahl@su.se; diez.molino@gmail.com

Introduction

How biotic and abiotic processes have shaped the evolution and diversity of life on our planet is a central question of the biological sciences. The advent of palaeogenomics provides opportunities to look directly into the past of species and ecosystems to understand their origin, evolution and even extinction. Unfortunately, ancient DNA (aDNA) is characterized by being highly fragmented, damaged and often contaminated by the degradation processes occurring after the death of an organism¹. Nevertheless, advances in laboratory methods and sequencing technologies have made it possible to obtain and analyse DNA from organisms that lived more than a million years ago (Ma)^{2,3}. Organism remains, such as bones or other tissues that have not undergone remineralization, are the most common sources of aDNA, but it can also be recovered from other sources such as fluid-preserved specimens, submerged wood, calcified deposits, coprolites and sediments^{4–6}. This wide variety of aDNA sources makes palaeogenomics a highly versatile tool with which to test a wide range of evolutionary and ecological hypotheses at different timescales.

The rate of ongoing biodiversity loss is alarming, and an increased extinction rate over the past few centuries has led to the claim that we are now undergoing a sixth mass extinction^{7–10}. However, cycles of biodiversity loss and recovery have occurred multiple times in Earth's history. Palaeogenomic methods now allow scientists to investigate the cause, duration and extent of previous large-scale biodiversity losses to better contextualize our situation today and forecast such trends. aDNA can be used to study past biodiversity change in response to climate shifts and human impact by reconstructing ancient ecosystems, species distributions and the genetic diversity of entire species or specific populations. Advances within the field of palaeogenomics, such as cheaper and more efficient sequencing techniques and laboratory methods that enable the recovery of shorter DNA fragments, have resulted in a wide variety of studies exploring central topics in biodiversity research. For example, high-resolution metabarcoding of ancient environmental DNA (the genetic material left by organisms in ancient environments) samples from Alaska recorded major changes in ecosystem composition for the past 15,000 years in response to climate change, a pattern that was not evident in pollen and macrofossil records¹¹. Similarly, temporal diversity of taxa inferred from ancient environmental DNA over climatic shifts throughout the Holocene enabled the identification of a long-standing diversity hotspot in northern Svalbard¹². Moreover, aDNA provides insight into the long-term consequences of severe bottlenecks¹³ as well as adaptive evolution¹⁴, for example, through research on the woolly mammoth (*Mammuthus primigenius*). Technological advances are enabling aDNA studies on increasingly large spatial and temporal scales. This increased scope offers opportunities to study biodiversity responses, at all three hierarchical levels (ecosystem, species and genetic diversity; see below) to direct human impact and a shifting climate, in line with future predictions of global warming. Such knowledge will be crucial for understanding how biodiversity dynamics are currently affected, and for predictions of the future consequences of increasing human impact.

In this Review, we discuss how palaeogenomics can be used to study biodiversity dynamics through time. We outline the opportunities and limitations of the use of aDNA to uncover past ecosystem, species and genetic biodiversity dynamics, and describe how such knowledge can aid our understanding of ongoing and future biodiversity losses. We focus on three time periods with different climatic, faunal and human activity events: the Early and Middle Pleistocene epoch, the Late Quaternary Period, and the industrial era (Fig. 1).

These time periods cover the entire window for which aDNA studies can currently be applied, and enable us to discuss biodiversity dynamics in response to climatic events, the global spread of humans, and the intensification of contemporary human impact related to activities beginning with the Industrial Revolution.

Biodiversity through time

The Convention on Biological Diversity (CBD) recognizes three hierarchical levels of biodiversity: ecosystem, species and genetic diversity (Fig. 2). Ecosystem biodiversity refers to the variety of ecosystems within a defined geographical location, including forests, grasslands, tundra and oceans. Within ecosystems, species diversity refers to the number and distribution of all species (hereafter referred to as taxa). Finally, genetic diversity is the variation of different alleles within and among populations of the same taxon. All three hierarchies of biodiversity are important for ecosystem function, as well as nature's contribution to people.

Biodiversity has not been static throughout Earth's history. Instead, it has changed through time in response to climate changes as well as other planetary and cosmic events, and is the product of the fundamental process of evolution. Five major timepoints throughout Earth's history stand out in terms of biodiversity loss, at each of which more than 75% of Earth's species are estimated to have gone extinct; the end Ordovician Period (445 Ma), the Late Devonian Period (372 Ma), the end Permian Period (252 Ma), the Late Triassic Period (201.3 Ma) and the Cretaceous–Tertiary (K–T) events (66 Ma)^{15,16}. These mass extinctions are well documented in the fossil record, but the ecological consequences are difficult to discern in detail because they all happened many millions of years ago. However, periods of extensive biodiversity loss and large-scale rearrangements of natural communities are not exclusive to these mass extinction events, and can be identified within the past few million years. The Pleistocene (~2.6 Ma to ~11.7 thousand years ago (ka)) was an epoch characterized by glacial periods interspersed by interglacial periods, which caused rapid shifts in community compositions and the distributions of taxa¹. During glacials, cold-adapted species could expand their ranges away from the poles, whereas warm-adapted species were confined to glacial refugia, and the opposite was true during warm interglacials¹⁷. The end of the Pleistocene culminated in widespread extinctions of megafauna caused by rapid climate warming and/or human expansion, beginning around 50 ka (refs. 18,19). During this period, around two-thirds of all megafauna (including woolly mammoths, woolly rhinoceroses, sabre-toothed cats and cave bears) are estimated to have gone extinct, resulting in substantial effects on ecosystem structure and functioning^{20,21}.

The present-day extent of biodiversity loss has not yet reached the level of previous mass extinctions, but current trends indicate that this could worsen. Around 25% of all known taxa are experiencing population declines so severe that they are at risk of extinction²². Moreover, 10% of genetic diversity could already have been lost in both threatened and non-threatened taxa²³, indicating that the adaptive potential – that is, the ability to evolve in response to changes in the surrounding environment²⁴ – of many populations is probably severely reduced and that the loss of biodiversity could have long-lasting consequences.

In our view, the climatic fluctuations and human expansions during the past 2.6 million years constitute a valuable framework within which it is possible to empirically investigate the causes and consequences of past biodiversity change across different hierarchical levels. Although most palaeogenomics research has focused on the past

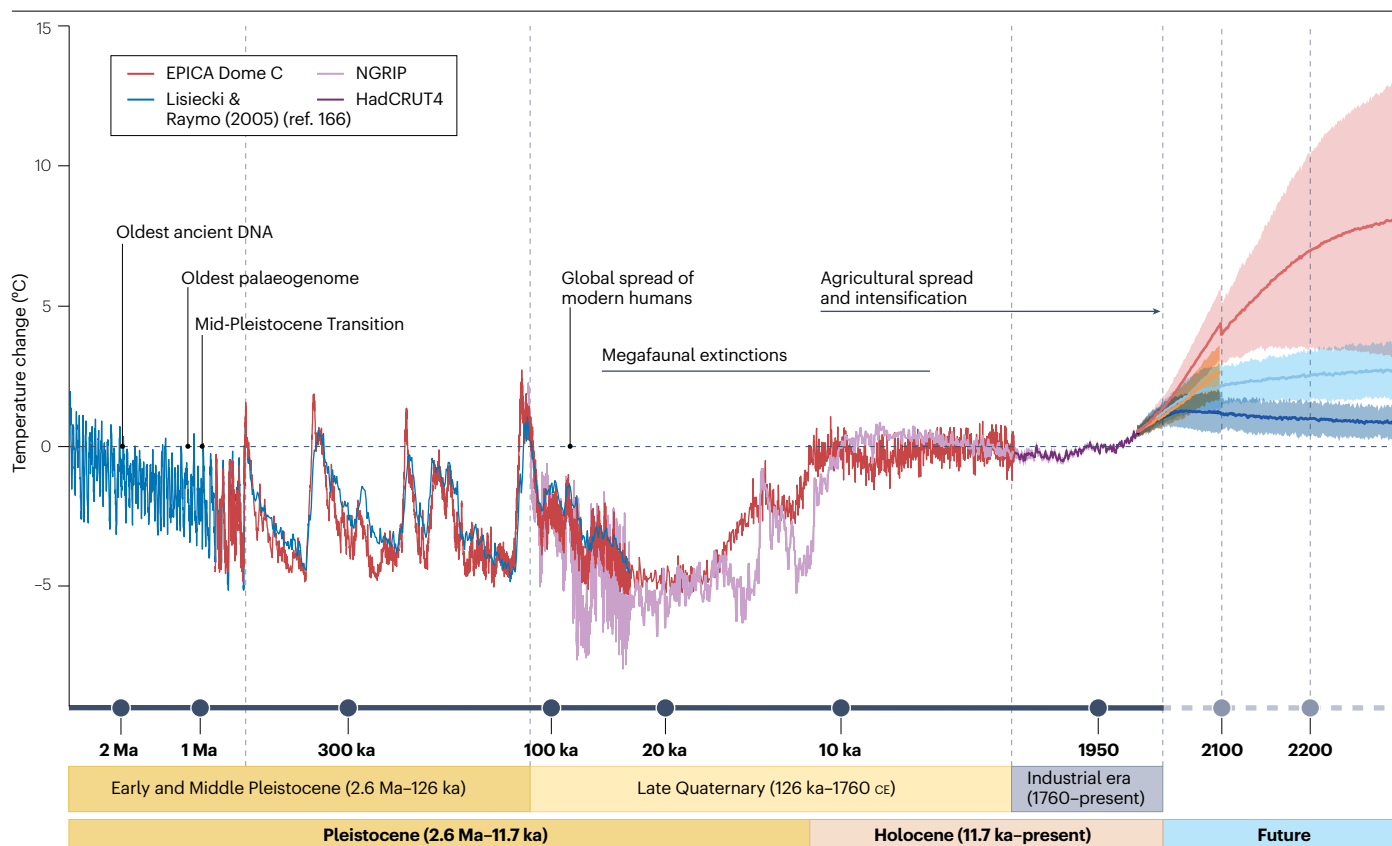


Fig. 1 | Temperature trends and key biodiversity events from the Quaternary accessible using palaeogenomics. Temperature estimates are based on the European Project for Ice Coring in Antarctica (EPICA; red line)¹⁶³, the North Greenland Ice Core Project (NGRIP; pink line)¹⁶⁴, the Hadley Centre and Climatic Research Unit Temperature (HadCRUT4; purple line)¹⁶⁵ datasets, and the LR04 stack of benthic $\delta^{18}\text{O}$ records (blue line)¹⁶⁶. Projections of near-future temperatures correspond to the four Representative Concentration Pathways (RCP2.6, RCP4.5, RCP6.0, and RCP8.5 in purple, blue, red and orange,

respectively) from the IPCC 2013 report¹⁶⁷. Major biodiversity events overlaid include the oldest sequenced ancient DNA (from ~2 million years ago (Ma)), the oldest sequenced palaeogenome (~1.2 Ma), the Mid-Pleistocene transition (~1 Ma), the global spread of modern humans (~70 thousand years ago (ka)), the megafaunal extinctions (~50–4 ka), and the spread and intensification of agricultural practices (12 ka to present). Figure adapted with permission from ref. 168, Proceedings of the National Academy of Sciences (PNAS).

50,000 years, technical advances now enable aDNA research to explore deeper time periods in increasing detail and at different taxonomic levels¹. Thus, we have an opportunity to better understand ongoing and future changes in biodiversity.

Palaeogenomics in biodiversity research

The field of palaeogenomics has made notable progress since the first aDNA was recovered in the 1980s²⁵. In this section, we present some of the advances made to date, remaining challenges, and discuss how palaeogenomic methods can be implemented in biodiversity research.

History and challenges

Inferences about changes in biodiversity and patterns of evolutionary history of taxa have primarily been based on the morphological identification of fossil remains and the assemblages in which they occur. The fossil record provides important clues about the evolution, richness and distribution of taxa in the past, but it is scarce and often taphonomically biased. For example, the fossilization process depends on multiple factors such as body size or geographical region²⁶,

and many organisms do not fossilize at all²⁷. Such biases potentially limit our ability to rely only on fossil remains when making precise inferences on past biodiversity. Complementary information can be gained by including aDNA in biodiversity research, because it provides new avenues to study the intra- and interspecific variation of taxa to reveal evolutionary processes that are difficult to study using fossil remains or present-day genetic data alone (Fig. 2).

Early aDNA studies were restricted by the available methodologies (namely polymerase chain reaction (PCR) in combination with Sanger sequencing), typically allowing the recovery of only fairly short fragments (100–500 base pairs (bp)) of the mitochondrial genome from periods limited to around 100,000 years ago^{25,28–30}. The field substantially expanded with the advent of next-generation sequencing and optimized laboratory protocols, reducing the need for targeted sequencing and enabling analysis of shorter fragments of DNA³¹. Although ultrashort fragments can be sequenced, it is more typical to limit analyses to those over >30 bp owing to the difficulty of aligning ultrashort fragments. Overall, next-generation sequencing technologies have advanced the possibilities of aDNA research, in terms of the

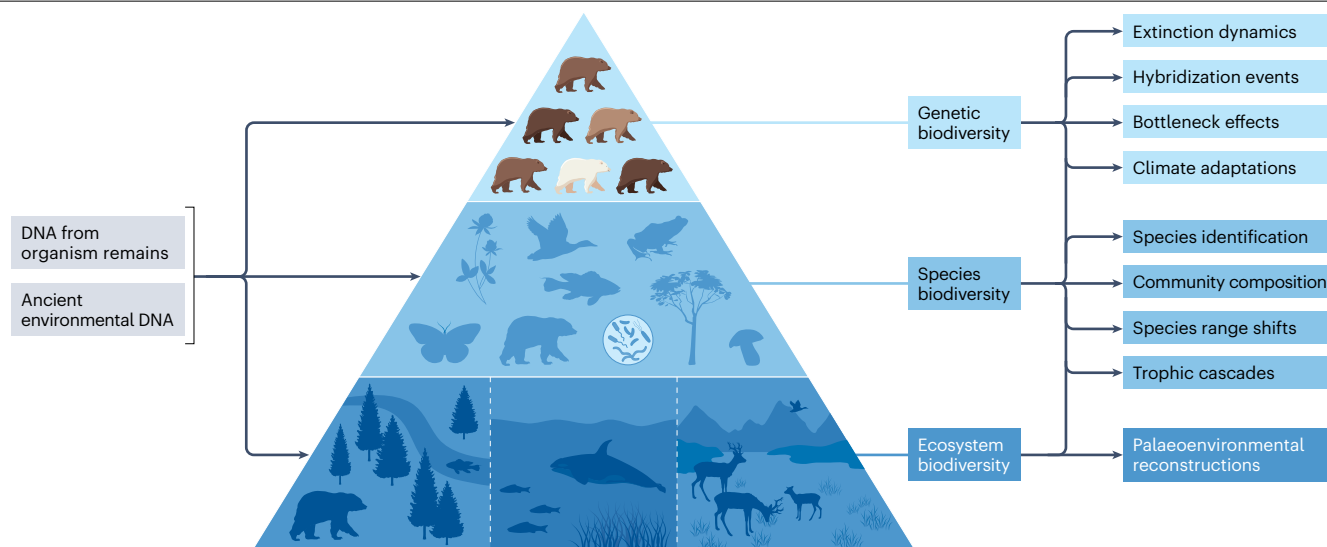


Fig. 2 | Palaeogenomic insights across the biodiversity hierarchy. The three core hierarchical levels of biodiversity are ecosystem, species and genetic diversity. Ecosystem biodiversity encompasses the variety of ecosystems, such as forests, oceans or grasslands within a given area. Species biodiversity refers to the number and distribution of species within a geographic location.

Genetic diversity is the variation and distribution of alleles within a species. Integrating genomic data extracted from organismal remains or ancient environmental samples (leftward of the pyramid) can contribute to a range of insights pertaining to aspects of genetic, species and ecosystem biodiversity (rightward of the pyramid).

proportion of the genome that can be recovered (nuclear as well as mitochondrial), and the depth of time and substrates from which aDNA can be retrieved. Today, technology allows for sequencing many genomes at the same time, thus enabling palaeogenomic analyses at the population or even community level³². Furthermore, under certain preservation conditions – typically cold and dry environments – whole genomes from as far back in time as the Middle Pleistocene and older (>1 Ma) can be recovered¹, providing clues about biodiversity change across large timescales. Since 2006, the application of genomic analyses in biodiversity research has gained increased attention thanks to their utility in monitoring biodiversity and assessing conservation strategies^{33–35}. aDNA adds to such research by providing a direct measure of past biodiversity, and of the causes and consequences of dynamics that have previously unfolded.

The field of palaeogenomics also faces some challenges. Important DNA fragmentation can occur fairly quickly after death with average fragment lengths of <100 bp reported^{36,37}, even in historical samples less than a century old³⁸. The reduced amounts of endogenous DNA preserved in a given sample results in a substantial contamination risk from exogenous DNA. Such contamination stems directly from the environment in which the sample is preserved or could be introduced during sample handling and laboratory analyses³⁹. Finally, the deamination of DNA, especially of cytosine, which is converted to uracil at the ends of DNA fragments, results in erroneous reading of DNA sequences⁴⁰. Notably, in palaeobiodiversity assessments, DNA damage and contamination can also lead to false-positive taxonomic assignments^{26,27}, or cause taxa to go undetected owing to poor DNA preservation (resulting in false negatives). To overcome these challenges, laboratory methods have advanced alongside sequencing technology to ensure that shorter fragments are maintained in the DNA-extraction process and that contamination is minimized during all steps⁴¹. Simultaneously, downstream bioinformatic approaches are

under constant improvement to optimize the mapping and analyses of short and damaged DNA fragments that are characteristic of aDNA.

Even under favourable conditions only small amounts of the original DNA remains. Degradation of DNA is particularly rapid in warm climates, limiting palaeogenomic studies in tropical and subtropical regions to timescales of centuries or at most a few thousand years^{42–44}. This global variation in degradation of DNA produces a bias in palaeogenomic studies: most published data comes from cold climate areas, but the most intense periods of human impact on biodiversity occurred globally⁴⁵. Furthermore, genetic identification of taxa depends on available reference genomes, which are currently lacking for many organisms. The absence of such reference genomes is arguably most critical when considering metagenomic or bulk analyses of aDNA, such as those obtained from ancient environmental DNA, because the taxonomic identifications derived are by necessity only as accurate or informative as the databases available to which we can compare the ancient environmental DNA data. Continued initiatives to sequence the DNA of life on Earth – for example, through the [Earth BioGenome Project](#) – are, however, ever increasing the availability of reference databases. Despite the challenges, the inclusion of palaeogenomics in assessments of past biodiversity allows us to produce a more complete view of biodiversity change at all three hierarchical levels.

Methods and applications

Palaeogenomic data can be recovered from diverse sources and used to study past dynamics at all levels of biodiversity. Ancient environmental samples can be obtained from sediments, ice cores and water, which contain mixed genetic traces of taxa derived from deposits such as plant seeds, wood and pollen, as well as animal faeces, hair and other tissues^{3,46,47}. Through sequencing of sedimentary aDNA (sedaDNA), past ecosystems can be reconstructed to fill gaps in the often highly incomplete fossil record. Additionally, sedaDNA can enable detection

and acquisition of genome-scale information for certain taxa in time and space, which can reveal species- and population-level genetic biodiversity dynamics^{48,49} in response to both natural and human-induced disturbances.

Three main methods are used to recover sedaDNA⁵⁰: shotgun metagenomics^{3,51}, metabarcoding^{52,53} and target enrichment^{54–56} (Fig. 3). The methods differ in terms of cost and the amount of genetic information recovered⁵⁰. Shotgun metagenomics is costly but can recover any genomic information present for taxa in a sample, whereas metabarcoding requires fewer resources but recovers only limited genomic data in the form of short barcode regions. However, shotgun metagenomics is currently more limited and biased than metabarcoding owing to the sparseness of reference databases for taxonomic identification⁵⁷. On the other hand, target enrichment uses baits or probes that bind, or hybridize, to genomic regions of interest to capture these specific regions^{58,59}. These regions of interest vary from single-sequence motifs to the entire genome, and can be used to efficiently increase the yield of genomic information from the selected

regions of the desired taxa. The high cost of capture baits often makes this method the most expensive of the three, but the sequencing effort required is substantially reduced and genomic information can be retrieved with a high degree of efficiency. Despite their individual limitations, these methods each provide a powerful tool for reconstructing assemblages of taxa at different times and locations, which can provide valuable data for assessments of ecosystem composition, diversity and connectivity over time.

aDNA can also be obtained directly from single-organism remains, such as bones and plant remains using shotgun metagenomics or target enrichment^{60–62}. Genetic analysis of remains can contribute to the correct taxonomic assignment of fossil taxa. For example, the extinct giant moa (*Dinornis* sp.) had three separate species described through morphological analyses, but aDNA analyses indicated that there were in reality only two taxa that each displayed sexual size dimorphism^{63,64}. Accurate identification of taxa is imperative for exploring processes affecting species diversity over time, such as range shifts, speciation events and extinctions. Lastly, aDNA analysis of remains offers a way

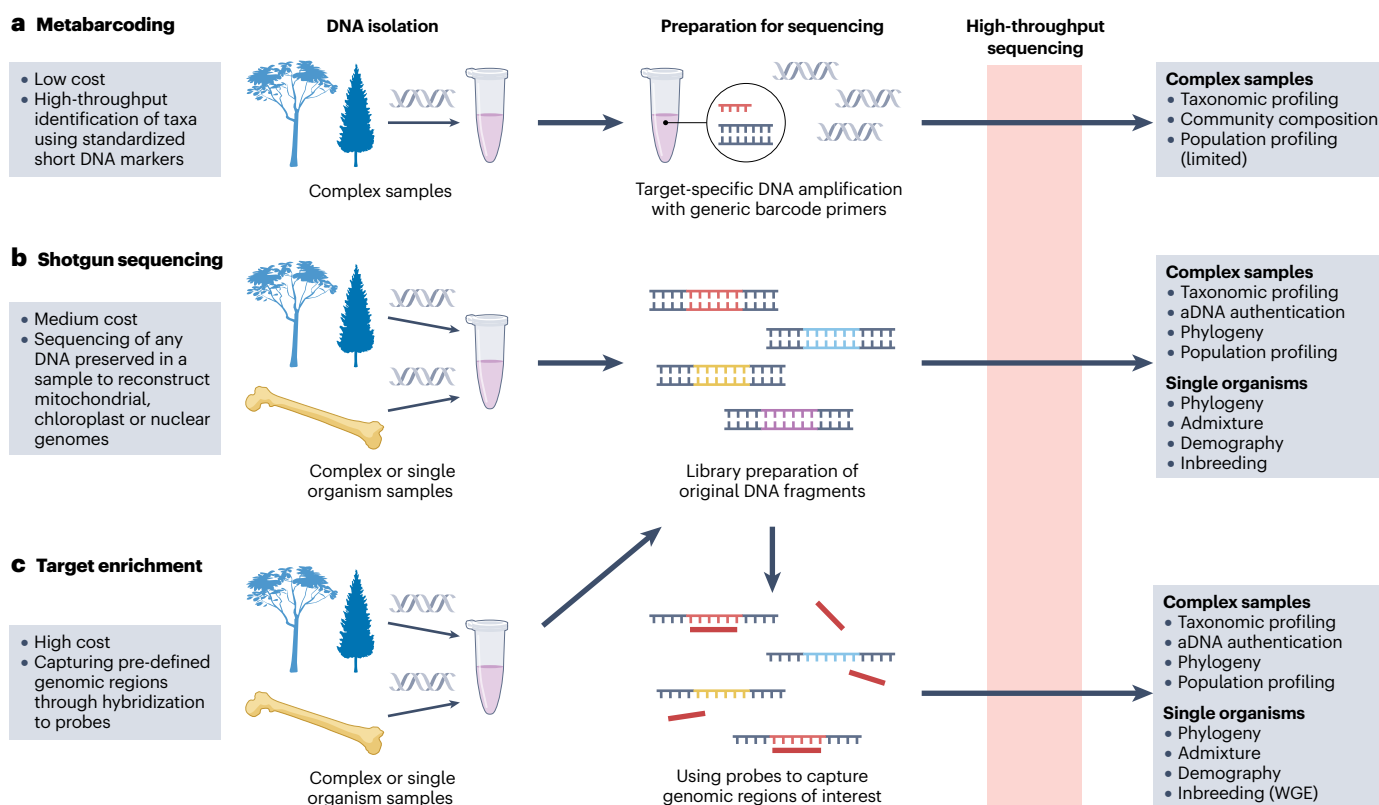


Fig. 3 | Simplified overview of common ancient DNA methods for investigating biodiversity. Grey boxes on the left show the relative cost and main goal of the approach. Grey boxes on the right show potential applications of output data that can be derived from each sample type using each approach. **a**, Metabarcoding approaches^{113,169} target DNA from complex samples by annealing primers to the barcode regions in mitochondrial, nuclear or plastid genomes. The barcoded regions should be variable enough to enable taxonomic assignments but flanked by conserved regions to allow for the use of universal primers for a taxonomically broad target group. High-throughput sequencing generates data capable of providing community-level insights. **b**, Shotgun metagenomics sequencing⁶⁰ can be applied to both single organisms and complex samples, individually sequencing all DNA fragments in the sample.

After sequencing, the fragments are aligned to a species target reference genome or a database of reference genomes either to reconstruct single-organism genomes or to identify sequences from multiple organisms in complex samples. **c**, Target enrichment^{58,59} can also be applied to single organisms and complex samples. Pre-designed probes are used to capture DNA fragments from specific genomic regions of interest, or even entire genomes (whole-genome enrichment (WGE)). The captured molecules are then retained for sequencing to increase their relative representation in the library and therefore the amount of sequencing data produced from them. Shotgun metagenomics and target-enrichment sequencing provide the highest-resolution data for investigating biodiversity in ancient systems.

to directly assess past genetic diversity. Tracking genetic diversity through time can reveal past processes that gave rise to the genetic structure of extant populations, such as hybridization events^{65–67}, genetic drift^{68,69} or population turnovers^{70–72}. Furthermore, the evolution of adaptive traits and the genetic processes of past extinctions can be studied with palaeogenomics, which is becoming increasingly important with the current rates of biodiversity loss.

Ecosystem dynamics

Ecosystems consist of biotic organisms and the abiotic components shaping their physical environment (for example, soil, sunlight and water) within a delimited geographical region⁷³. These diverse combinations of biotic organisms and abiotic factors are what define different ecosystems, such as coniferous, deciduous or mixed forest ecosystems. For any defined geographical region, the diversity of ecosystems contributes to multiple regulating, provisioning and supportive services crucial to the well-being of both humans and other organisms. In general, palaeogenomics has not yet been used to investigate the diversity of ecosystems within specific geographical regions. One reason for this

is that it can be challenging to resolve whether results (for example, from sedaDNA studies) represent single or several ecosystems (Fig. 4). This resolution is something that could be addressed in future studies, for example, through systematic sampling of palaeoenvironmental samples aimed at discovering different ecosystems within the same region. Despite these challenges, sedaDNA research has provided a first glimpse of the potential to study past ecosystem diversity.

Early and Middle Pleistocene

The Early and Middle Pleistocene (2.6 Ma to 126 ka) were characterized by climate oscillations and a shift to longer glacial intervals, from cycles of ~40 ka to ~100 ka occurring around 1.2 to 0.7 Ma (ref. 74). This period contained both warmer, colder, drier and wetter climates than present, which provides unique opportunities to investigate ecosystem dynamics in response to climate changes that were not driven by humans.

As of the end of 2024, the oldest DNA that has been recovered is from a ~2-million-year-old (Early Pleistocene) geological deposit in Northern Greenland³. This revealed a unique species composition that has no known modern analogue, consisting of a highly diverse and

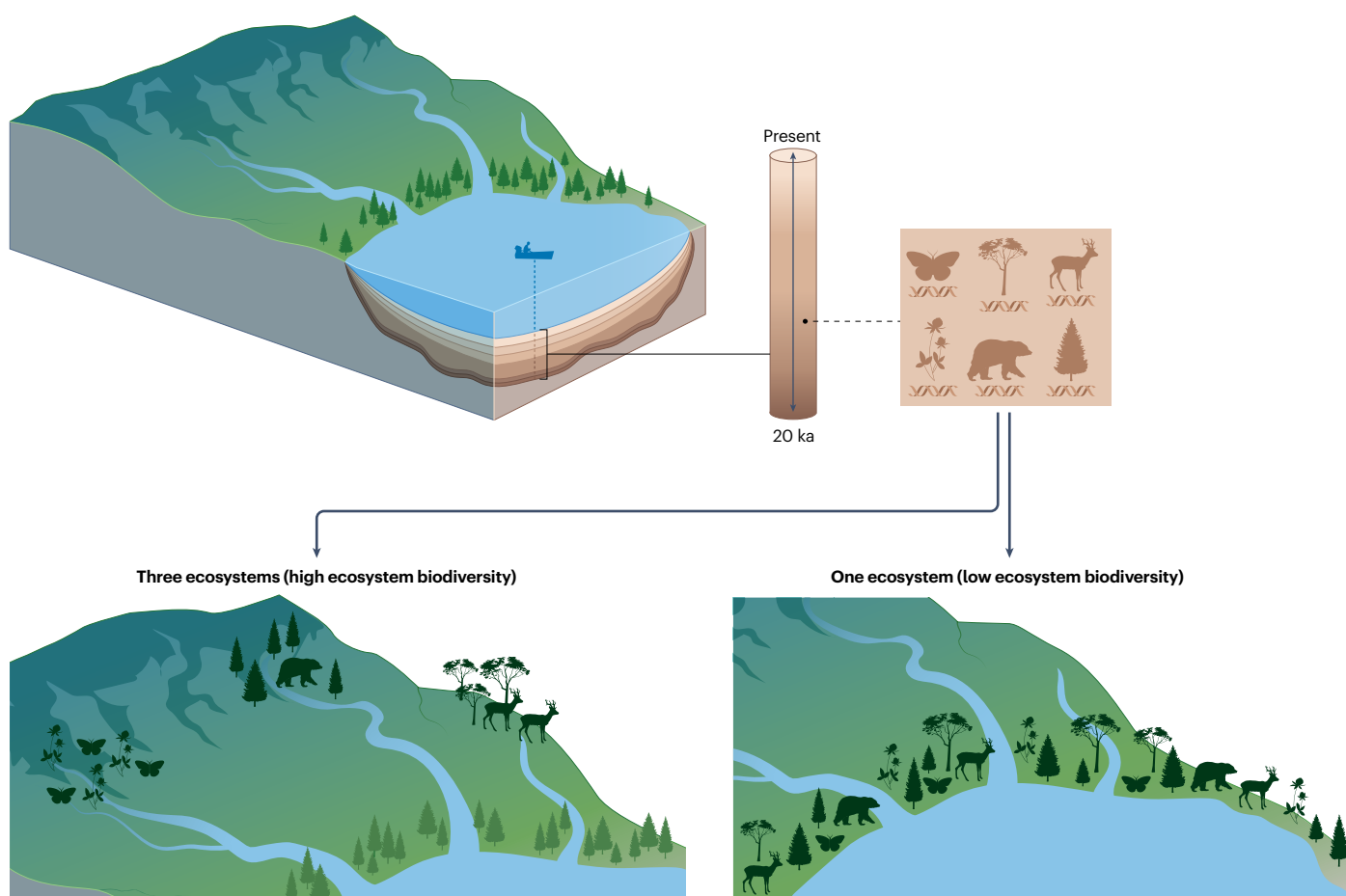


Fig. 4 | Challenges of studying ecosystem biodiversity with ancient DNA. Ancient environmental DNA is useful for exploring past biodiversity dynamics at the ecosystem level, yet it has challenges. For example, it is not possible to resolve whether ancient environmental DNA taken from a hypothetical lake sediment core represents the biodiversity of a single ecosystem or of several ecosystems. One scenario illustrates high ecosystem diversity within a defined geographical region, whereby three distinct ecosystems contribute environmental genomic

data to a single central lake. The second scenario illustrates a defined geographical region of the same size with high species diversity but low ecosystem diversity, because all taxa are from the same mixed forest on the lakeside. Both scenarios would yield a similar ancient environmental DNA signal, making it difficult to resolve whether the region surrounding the lake has high or low ecosystem biodiversity. Figure adapted with permission from ref. 170, John Wiley & Sons.

productive open-forested ecotone that stands in contrast to the less productive and less diverse Arctic desert found in northern Greenland today. The presence of warm-adapted boreal and aquatic taxa intermixed with cold-adapted taxa suggests that Greenland had a mosaic of different ecosystems at this time in the Early Pleistocene, and that these past species were adapted to this extreme insular regime.

Later interglacials, after the Early Pleistocene, featured different latitudinal movements of plants in response to climate change across the Arctic, particularly the expansion of boreal taxa during warmer climates^{75–77}. During the last interglacial (129–116 ka), the expansion of boreal taxa is believed to have amplified warming from summer insolation in the Arctic through vegetation-related feedbacks, causing high-latitude landscapes to be ice-free for longer and allowing for even more substantial vegetation shifts⁷⁶. Climate projections by the [Intergovernmental Panel on Climate Change](#) estimate that average temperatures could exceed 4 °C above pre-industrial levels by the end of the twenty-first century⁷⁸, and trends towards greening are already occurring in the Arctic⁷⁹.

Reconstructing past ecosystems during previous warmer-than-present climates could therefore be informative for future changes in Arctic ecosystem diversity, and the dynamics that could result from increased greening. For example, analyses of sedaDNA, which provide a highly local signal of vegetation change⁸⁰, could be used to compare the nature and process of vegetational succession events across multiple sites that took place at the onset of previous interglacials with such events of the early Holocene, all at a high temporal resolution. These reconstructions could inform not only how representative past interglacial warming is of present-day climate change, but also, given that sedaDNA can provide a high taxonomic resolution, whether expansions of particular boreal taxa foreshadow amplified insolation-driven warming. Moreover, sedaDNA analysis of previous interglacials that lasted longer than the current Holocene (that is, >12 thousand years) could provide clues on time-lags in biodiversity change that have yet to materialize.

Late Quaternary

Between ~126 ka and 1760 CE, here referred to as the Late Quaternary, large-scale environmental change, the global spread of humans and the intensification of human practices have jointly influenced biodiversity¹⁹. In temperate and colder climates⁴⁵, much of this period is accessible for aDNA analyses from both sediments and organism remains, providing an opportunity to study biodiversity change through time both in the absence and in the presence of pre-industrial human impact.

The Pleistocene–Holocene transition, which occurred 11.7 ka, is of particular interest for biodiversity research, involving large-scale turnovers of species across all trophic levels and the widespread extinctions of megafauna⁸¹. In the Arctic, terrestrial ecosystems shifted dramatically following the Last Glacial Maximum (20–26 ka), when the glaciers retreated and temperature and moisture levels increased. Specifically, sedaDNA has contributed to understanding the shift from previously open landscapes dominated by cold-adapted steppe taxa to a mosaic of steppe, boreal and aquatic taxa^{52,82}.

The widespread ecosystem changes of the Pleistocene–Holocene transition also caused bottom-up trophic cascades and the well-known widespread extinctions of megafauna in large parts of the world, probably exacerbated by human activity^{83,84}. Following this transition phase, the buildup of ecosystems in the Arctic demonstrated a continuing increase in plant species diversity throughout the first half of the Holocene⁸⁵, whereas ecological trait and functional diversity

stabilized after rapid changes during the first millennia⁸⁶. The aDNA analysis of bulk bone complements the fossil record and enables tracking of communities of both small and large fauna during the different warming phases of the Holocene, including fish, birds and mammals from both marine and terrestrial environments⁸⁷. Despite dramatic turnovers in much of the world, regional differences are apparent and ecosystem composition and diversity remained relatively stable in, for example, both northwest and central Siberia⁸² and in the polar Urals⁸⁸ as the climate shifted. Areas that displayed high ecosystem resilience through periods of warming in the past are important to identify, as these may act as future refugia for cold-adapted species.

Traces of human impact become evident in Holocene fossil and aDNA records as sampling approaches the modern day. In the European Alps, an abrupt shift with novel species compositions occurred as forest ecosystems were almost completely replaced with open heathlands and grazed lands resulting from pastoral farming activities^{89,90}. A similar pattern has been observed in southeastern Norway, where ecosystem composition prior to human impact had remained stable throughout the Holocene but shifted to a more diverse, anthropogenic environment as agriculture was introduced to the region and local farms were established⁹¹ (Fig. 5).

So far, the majority of palaeogenomic reconstructions of ecosystem diversity in the Late Quaternary have been restricted to the Arctic, where low temperatures are optimal for DNA preservation⁹². However, Holocene sedaDNA recovered from alpine environments in Africa⁹³ and Asia⁹⁴ highlight the potential for future studies in other parts of the world. With advances in shotgun metagenomics sequencing, studies on lowland tropical ecosystems also become tenable, as demonstrated by the 2,200-year-old sedaDNA data from Uganda⁴⁴. Nevertheless, in these regions especially, effort is needed to increase the generation of reference databases that are essential for inferring assemblages from sedaDNA data. Given that the majority of Earth's biodiversity is concentrated in tropical areas, and many ecosystems in these areas are threatened⁹⁵, studies on how such ecosystems have responded to climate change and human impact over time should be highly informative and valuable for proposing effective conservation actions.

Industrial era

The beginning of industrialization in the 1700s initiated the large-scale human impact on biodiversity seen in recent centuries^{96,97}. Human practices such as agriculture, burning and deforestation have completely altered landscapes and community compositions⁹⁸. Moreover, the introduction of invasive species, overhunting, and the spread of infectious diseases related to human movement have caused widespread extinctions⁹⁹. aDNA provides an excellent opportunity to study the effect of human pressures on ecosystem biodiversity, but its full potential remains to be explored.

Importantly, most palaeogenomic studies of human impact on ecosystem diversity over the past few centuries focus on dynamics within single ecosystems, rather than ecosystem diversity (such as the variation in and number of ecosystems within a geographical area). For example, sedaDNA has been used to reconstruct forest ecosystems before and after the start of agricultural practices in the French Alps and in Norway^{90,91}, and to investigate changes in single lake ecosystems in response to eutrophication over the past few centuries^{100–102}. However, the intensified human impact that started during the Industrial Revolution a little less than 300 years ago caused biodiversity changes on ecosystem scales⁹⁶, and many species rely on more than one ecosystem for reproduction and survival. For example,

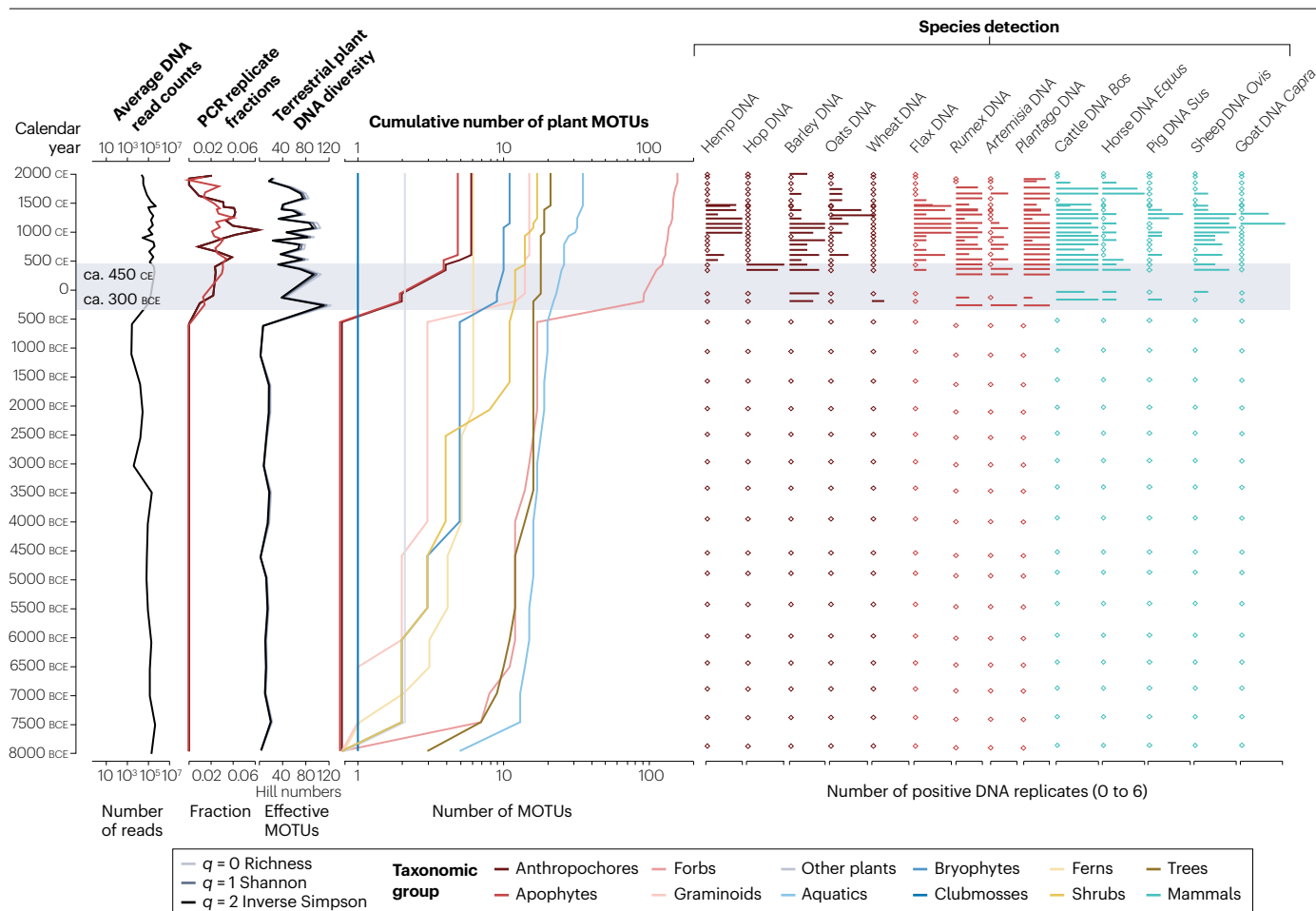


Fig. 5 | Ancient environmental DNA records of human disturbance in sediment cores from Lake Ljøgottjern, Norway. From left to right: average DNA read counts (black line); PCR replicate fractions for Anthropochores (brown line) and Apophytes (red line); terrestrial plant DNA diversity, as determined by Hill numbers. q values represent effective molecular operational taxonomic units (MOTUs) in the form of species richness ($q = 0$; light grey line), Shannon index ($q = 1$; dark grey line) and inverse Simpson index ($q = 2$; black line); cumulative number of plant MOTUs for a range of taxonomic groups (coloured lines); presence and absence of DNA from Anthropochores (brown elements),

Apophytes (red elements) and mammalian livestock (cyan elements). Dots represent absences, lines indicate number of positive DNA replicates per sample (up to $n = 6$). All results are presented over a time span of approximately 10,000 years. Grey shading indicates a time period of considerable cumulative increases in plant and animal taxa detection. This period coincides with increased human agricultural practices and pastoralism determined by local archaeological evidence. CE, Common Era; BCE, Before Common Era. Figure adapted from ref. 91, CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0/>).

migratory species utilize different ecosystems depending on life stage or season, and contribute to ecosystem function and services with their movements¹⁰³. Such species are threatened by processes that have intensified in the past few centuries, including habitat loss, overexploitation and climate change¹⁰³, and the recent global reduction of terrestrial mammalian movements has been attributed to human impact¹⁰⁴. This reliance on multiple ecosystems emphasizes the importance of monitoring ecosystem biodiversity change at an ecosystem level and extending geographical sampling in historical assessments.

Ancient environmental DNA studies have proved useful for ecosystem reconstructions, given that they can detect a broad range of organisms with high taxonomic resolution, as exemplified by reconstructions of glacial and interglacial communities^{77,82,84}.

Environmental DNA shows great potential for reconstructing changes in biodiversity across different ecosystems in response to increasing human pressures over the past few centuries, for example, in aquatic ecosystems¹⁰⁵. Broad geographical sampling using ancient environmental DNA suggests a trend of homogenization in aquatic communities over the past century^{106,107} – a pattern that would be missed in single lake studies.

To fully capture ecosystem biodiversity changes and potential consequences, studies would ideally include sediment samples that represent a wider variety of ecosystems across regions, and use optimized sequencing techniques that capture as many taxa as possible. Currently, most such studies use metabarcoding, which provides inventories of taxa of the targeted phylogenetic group and are highly efficient in terms of laboratory processing, financial cost and computational

time required. However, further growth of genomic reference databases and development of analytical procedures will potentially enable a more efficient use of shotgun metagenomics sequencing^{3,51}. These methods are being applied to reconstruct ecosystem biodiversity further back in time^{3,82,108}. Despite limitations with stratigraphy and dating within the industrial era, the dramatically increasing rates of biodiversity loss in recent centuries⁹⁸ urgently call for an increase in the usability of these methods on centennial scales as well.

Species diversity

Species biodiversity can be assessed by the number of species within an area (species richness) and their abundance (species evenness), but also based on differences in functional traits (functional diversity) or phylogenetic distances (phylogenetic diversity)¹⁰⁹. High species diversity contributes to resilient, stable and productive ecosystems, and is maintained by a balance between speciation and extinction^{110,111}. Sometimes, however, the extinction of taxa exceeds speciation, with consequences for entire ecosystems. Here we present palaeogenomic insight into how species-diversity dynamics have changed throughout the Quaternary.

Early and Middle Pleistocene

The diversity of species is influenced by evolutionary processes that occur over long periods of time, such as speciation, migration and extinction. These processes fluctuate in response to changes in the environment, which can be demonstrated with aDNA as the presence or absence of species at a site during certain time periods. In the Early Pleistocene, records from northern Greenland include species that are presently confined to southern latitudes, such as cedar trees (*Thuja* sp.), corals (Merulinidae), and horseshoe crab (Limulidae), as well as the extinct mastodon (Mammutidae)³. Phylogenetic placement of key taxa in this record indicated that these taxa were likely to be ancestral and therefore could potentially represent extinct lineages. Genomic studies on Early and Middle Pleistocene materials, also known as deep-time palaeogenomics¹, show great potential for reconstructing faunal assemblages, and can be used to detect ranges of extinct taxa and past ranges of extant taxa in response to climate change.

Deep-time palaeogenomics can also be used to study community composition dynamics and trophic interactions during past major climatic shifts. For example, faunal records from marine isotope stage (MIS) 17 (~712–676 ka) in East Siberia revealed an interglacial community consisting of grassland plants such as Poaceae and Asteraceae, and megafauna such as bovids (Bovidae)⁷⁷. However, in MIS 7 (~243–191 ka) the abundance of megafauna decreased, and the number of forest-dwelling small mammals and forest-specific insects increased⁷⁷. This shift could suggest a trophic link between megafauna and open grasslands, in which megafauna contributed to keeping the landscape open through grazing and trampling. Drastic community composition changes have also been recorded in the marine realm during the transition from the Middle to Late Pleistocene. For example, deglaciation has been correlated with a marked increase in diatom abundance, which increased from ~10% to ~50% of all eukaryotes, and open ocean diatoms increased at the expense of sea-ice diatoms¹⁰⁸. Such community composition changes undoubtedly affected marine ecosystems because diatoms are important primary producers¹¹². Although only a few DNA fragments could be recovered from the oldest three sediment samples – dating to ~2.5–0.7 Ma, 1 Ma and 417 ka – this information is important to understand which species persist through time, and which are lost. For example, certain diatom

taxa seem to have remained quite stable through time, such as Mediophyceae, whereas others have disappeared, such as Eunotogramma¹⁰⁸. Observations of community structure responses to climate change could increase our understanding of how ecosystem composition might change in a warmer future, and identify ecosystem functions at risk.

Overall, using palaeogenomics to study changes in species diversity on a multi-million-year timescale has the potential to address several fundamental biological questions, especially given the power of aDNA in discovering previously unknown and long-extinct taxa and lineages². Thus, palaeogenomics could be used to investigate the rates of extinction across multiple glacial cycles, as well as to what extent range expansions have been driven by past sea-level dynamics.

Late Quaternary

Global climate oscillations and increasing human pressure in the Late Quaternary have substantially affected species diversity and distributions. At the species level, range shifts can be detected as the presence or absence of certain taxa within a geographical area. For example, northern bog lemmings (*Synaptomus cooperi*) and northern grasshopper mice (*Onychomys leucogaster*) appeared in Last Glacial Maximum layers from Texas, USA, and disappeared as the climate became warmer⁸⁴. Similarly, big brown bats (*Eptesicus fuscus*), which require low temperatures to hibernate, disappeared in the warm Bølling–Allerød interstadial⁸⁴. These findings suggest that range shifts throughout the Pleistocene were largely climate-driven. DNA-based bulk-bone reconstruction of faunal assemblages from ~46 ka in southwest Australia¹¹³ identified two extinct tree-kangaroo taxa (*Bohru* sp.), although tree kangaroos are currently found much further north today. These studies highlight the potential of palaeogenomics to detect species and thus to investigate their range shifts in response to different pressures in the past.

Range shifts into new environments can initiate adaptive responses and fast radiation of taxa, and ultimately an increase in species richness. A remarkable example has been identified in Caribbean rodents, where aDNA analysis revealed that both extant and extinct taxa, together displaying broad morphological and ecological variation, originated from a single lineage and colonization event¹¹⁴. Studies such as these can increase our understanding of speciation and the genetic and environmental factors underlying rapid adaptation. The Late Quaternary provides an excellent opportunity to investigate such dynamics because many species shifted their ranges during this period^{84,115–117}.

In the latter half of the last glacial period (starting ~50 ka), species diversity in many parts of the world was severely affected by the widespread extinction of megafauna. aDNA studies have been key to understanding the consequences that followed. Trophic collapses resulting from the loss of megafauna are reported from different parts of the world during this time^{77,84,118}, inferred by the presence or absence of taxa. Furthermore, aDNA studies have been important for inferring the timing of extinction events, and have even led to the reassignment of the timing of extinction events from the Late Pleistocene to the Holocene^{119,120}. For example, mammoth aDNA was detected in ~5,600-year-old sediments from St Paul Island, which, together with other lines of evidence, demonstrate that mammoths survived there for thousands of years after mainland populations went extinct¹²⁰. Findings such as these can help to identify the underlying causes of extinctions, and how taxa at different trophic levels can be affected by the loss of keystone taxa.

Industrial era

Humans have contributed to an accelerated loss of species diversity⁹⁸. The causes of these losses are almost exclusively anthropogenic, and include overhunting, human land use and the introduction of invasive species^{95,98,121}. Although these processes have intensified in the last few centuries, the consequences for species diversity and composition is evident in genetic studies using preserved material from historical sources dating back to around 300 years ago (historical DNA).

Many species extinctions over the past few hundred years can be attributed to overhunting and land use by humans. One well known example is the thylacine (*Thylacinus cynocephalus*), which went extinct in 1936 following overhunting after having been confined to Tasmania¹²². Similarly, the passenger pigeon (*Ectopistes migratorius*), great auk (*Pinguinus impennis*), and Steller's sea cow (*Hydrodamalis gigas*) have all been driven to extinction over the past few centuries owing to direct human persecution or habitat loss caused by human activity^{123–125}. Human impact has also altered community compositions, with important consequences for ecosystems. For example, sedaDNA studies show that in lacustrine and brackish systems, eutrophication related to human land use has caused shifts in the abundance and composition of primary producers^{101,126,127}, with effects on the entire food web. These shifts include taxon-specific responses to increasing temperatures and phosphorus levels, and irreversible turnovers of taxa^{97,100}. This non-return trajectory over time has important implications for the potential future resilience of lakes that experienced eutrophication, which might come into play under current global warming. Non-reversibility of community composition has been documented in Lake Constance in southern Germany after its re-oligotrophication following decades of anthropogenic eutrophication¹⁰², based on metabarcoding data from multiple taxonomic groups. Furthermore, irreversible change in the twentieth century following centuries of stability has been identified in the marine Bay of Brest in northwestern France, based on sediment cores covering more than a thousand years. The cumulative effect of World War II and agricultural pollution appears to have irreversibly altered protist communities from the second half of the twentieth century onward¹²⁸.

aDNA and sedaDNA can also help us to explore the timing and consequences of the introduction of invasive species. For example, historical DNA from ship ballast tanks and marine port sediments from Australia indicate that invasive and harmful microalgal taxa have increased over the past 30 years, and are dispersed by ships¹²⁹. On a sub-Antarctic island, the effect of invasive rabbits introduced in the nineteenth century was correlated with plant community composition changes, and a marked decrease in a keystone plant species¹³⁰.

The widespread extinction of species, rapid turnovers of taxa, and trophic cascades that have occurred in the past few centuries testify to the large-scale effect that humans have exerted on species biodiversity in the industrial era. Historical palaeogenomics demonstrates the potential for application to various research areas to investigate the processes behind recent species biodiversity declines. Despite the relative ease with which DNA can be recovered from specimens and sediments that are only a few hundred years old, there are very few palaeogenomic studies that have addressed questions related to the effect of industrialization on species diversity. Examples of potential research avenues include using sedaDNA to study the cascading effect of keystone species extinctions, changes in the abundance of taxa and species composition in response to fire regimes, and the timing, rate and extent of the introduction of novel diseases. These types of studies could contribute to our understanding of the current widespread biodiversity loss^{131,132}.

Genetic variation

Genetic diversity constitutes the substrate for evolutionary change and adaptation to new environments. However, the process of changes in genetic variation can be difficult to study using modern data alone, as these have taken place in response to past evolutionary events happening over both short and long periods of time. In this section, we discuss how palaeogenomics can contribute to our understanding of biodiversity through the analysis of genetic diversity dynamics through time.

Early and Middle Pleistocene

The most direct way to use palaeogenomics to assess genetic variation within a species is to quantify individual heterozygosity within multiple individuals¹³. However, the high degradation in samples that are hundreds of thousands and up to millions of years old makes it challenging, albeit in principle not impossible, to recover genomes of sufficient coverage to assess individual heterozygosity¹³³. An alternative approach when dealing with highly degraded samples is to generate low-coverage genomic data from multiple individuals and estimate the population-level genetic diversity based on allele frequencies. Although this approach is a more resource-efficient alternative, it has its own challenge in that diversity estimates could be biased if samples are from different time points¹³⁴. Should there be a sampling bias, the analyses would need to be corrected for the effect of genetic drift and de novo mutations.

Despite these challenges, the recovery and analyses of genomes from Early and Middle Pleistocene samples demonstrates potential utility^{2,135–137}. For example, the analysis of million-year-old mammoth palaeogenomes highlights a surprisingly high genetic diversity in Early Pleistocene Siberian mammoths, which belonged to two highly differentiated evolutionary lineages². This high diversity shaped the evolution of later mammoth taxa, with the Columbian mammoth (*Mammuthus columbi*) representing a mosaic of these two evolutionary lineages. Furthermore, a 360-thousand-year-old cave bear (*Ursus kudarensis*) palaeogenome was used to investigate the timings of when different bear lineages diverged, with this divergence linked to a major climatic shift in the Middle Pleistocene¹³⁶.

Genetic diversity can be studied on million-year timescales, which allows central questions in evolutionary genomics to be investigated. A key use case of this type of deep-time palaeogenomics approach could be to examine how much time it takes species to recover their genetic diversity following severe bottlenecks, a question of great relevance for conservation biology today. Moreover, Early and Middle Pleistocene genomes could provide a way to investigate the rate of adaptive genetic change, as recently illustrated in a study on woolly mammoths¹⁴. A broader application of this approach, encompassing multiple taxa to assess the rate of adaptive evolution in general and in response to past environmental change, could provide novel insights into how much time it will take present-day species to adapt to pollutants as well as to ongoing climate change.

Late Quaternary

The climatic changes during the Late Pleistocene and the onset of the Holocene caused widespread population declines, which led to losses of genetic diversity and the consequent reduction in adaptive potential as well as, for many species, their extinction. Genetic diversity can either be lost on large scales, such as the disappearance of entire clades, or on smaller scales, such as the loss of genetic variants in populations or individuals within populations. Palaeogenomics can be used

to detect both large- and small-scale losses of genetic diversity, and help to disentangle the relative roles that changes in climate and other factors, such as the increasing effect of human activity, have had in causing these losses. For example, small mammals such as collared lemmings (*Dicrostonyx torquatus*) and narrow-headed voles (*Stenocranius gregalis*) experienced rapid genetic turnovers and a considerable loss of genetic diversity during this period (Fig. 6). Given that humans are unlikely to have had a marked direct effect on small rodent dynamics, these findings illustrate the importance of past climate change as a driver for biodiversity change^{137,138}.

aDNA analyses of Late Quaternary genetic diversity can also provide information on population movements, including migration events¹³⁹, as well as identifying likely source populations for such movements^{140,141}. A further advantage of using aDNA to explore genetic diversity is that it can unmask past hybridization events between species, such as an admixture event that occurred between polar bears (*Ursus maritimus*) and brown bears (*Ursus arctos*) during the last interglacial, as their ranges shifted and started overlapping⁶⁷. Signatures of this introgression are still evident in modern brown bear genomes, but absent in polar bears, suggesting that admixed ancestry might have had a fitness cost in polar bears⁶⁷. Many extant species probably derive part of their ancestry from such past admixture events, and, for some, admixture could incur a substantial fitness cost.

aDNA can also provide insights into the genomic processes leading up to extinction through inferences on the interplay between demography and changes in genetic diversity over time. For example, low genetic variation and increased inbreeding was observed in the last surviving woolly mammoth population on Wrangel island when compared with ancestral and large mainland populations, which probably contributed to a loss of adaptive potential and eventually its extinction¹⁴². By contrast, woolly rhinoceros showed no signs of decreased heterozygosity or increased inbreeding shortly before its extinction, suggesting that its extinction might have happened more suddenly¹⁴³. Studies focused on generating palaeogenomic data on the extinction process of Late Quaternary species can provide clues on the temporal development of such processes, and therefore have the potential to be useful for conservation and management programmes for species currently in decline.

Industrial era

Although it is difficult to distinguish the roles of human impact and of climate change in Pleistocene extinctions, there are numerous examples of humans directly causing species declines in the past century^{144,145}. Historical DNA can provide insight into how human actions have contributed to these processes by enabling the comparison of genetic diversity before and after such declines¹⁴⁶.

Overexploitation is one of the most common causes of recent species extinctions, especially on islands^{147–149}. Understanding genomic processes during extinction is important for inferring conservation actions, and aDNA is a valuable tool for that purpose. For example, aDNA has been used to reconstruct demographic histories of species prior to extinction. Demographic reconstructions based on data primarily from museum specimens of the thylacine¹²², great auk¹²⁴, and endemic Australian rodents¹⁵⁰ all revealed no drastic decline in genetic diversity prior to extinction, indicating that extinction was a rapid process. The thylacine had a history of declining genetic diversity further back in time, throughout the Holocene, increasing its vulnerability to hunting pressures¹²². However, great auks and Australian rodent populations displayed high genetic diversity prior to extinction^{124,150}.

The passenger pigeon also went extinct as a consequence of overhunting, even though the population was large and widespread¹²³. aDNA studies of museum specimens have revealed that passenger pigeons had surprisingly low genetic diversity compared to their population size, with negative effects on the population as it declined¹²³. In these cases, the genetic history inferred from aDNA has revealed important clues to the underlying causes of species extinctions. The extinction process can be rapid and is not always preceded by low genetic diversity or caused by small and restricted populations. Furthermore, it has become evident that genetic diversity is important for species survival, but that genetic diversity alone cannot save a species from extinction if it is subjected to overexploitation or if its range is highly restricted.

Beyond exploring genomic processes in extinct species, aDNA analyses can also be applied to understand the effect of ongoing population declines. Small populations are vulnerable to genomic erosion, owing to loss of heterozygosity, increased inbreeding and the accumulation of harmful mutations¹⁴⁶. Eventually, these processes can cause the extinction of species, or have long-term effects on the survival of populations even after recovery. Emerging uses of aDNA include exploring the genetic diversity of species subjected to human-induced bottlenecks, as contemporary events can be overshadowed by past processes¹⁴⁶. For example, aDNA studies clearly demonstrate that

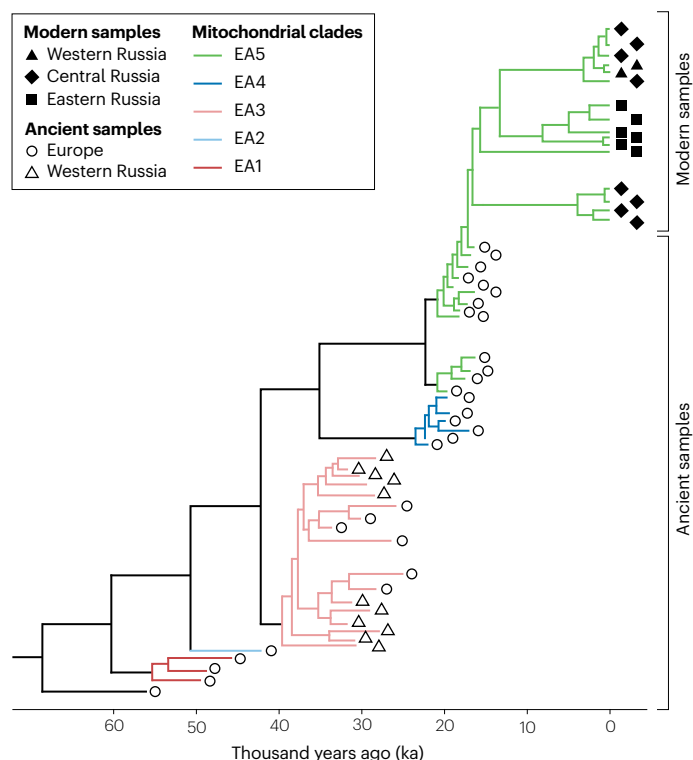


Fig. 6 | Ancient DNA reveals genetic turnover in the collared lemming during the Late Quaternary. Palaeogenomic analysis of over 300 fossil remains of collared lemming (*Dicrostonyx torquatus*) spanning the past 50,000 years records five sequential mitochondrial genetic clades (different coloured lines). Symbols denote locations, with filled symbols representing modern samples and unfilled symbols representing ancient samples. All modern samples belong to a single mitochondrial clade. The extant populations of collared lemming represent only a small fraction of their late Quaternary diversity. Figure adapted with permission from ref. 70, John Wiley & Sons.

Glossary

Baits

Baits or probes for target enrichment are made of short biotinylated single-stranded DNA or RNA molecules that bind to DNA fragments from specific genomic regions, which are then separated from the remaining non-target DNA molecules.

Bulk bone

A collection of morphologically unidentifiable bone fragments collected from a single stratigraphic layer in a cave, for example.

Endogenous DNA

The fraction of DNA in a sample that derives from the organism(s) under study.

Exogenous DNA

The fraction of DNA in a sample that derives from organisms other than the one(s) under study; this can be introduced by the environment in

which the sample is preserved, during handling of the sample, or by laboratory preparations.

Glacial periods

Intervals of time characterized by cold temperatures, the formation of extensive continental ice sheets and reduced sea levels.

Glacial refugia

Geographical areas with warmer climates that provide suitable living conditions for warm-adapted species during glacial periods.

Interglacial periods

Intervals of time between glacial periods characterized by warm temperatures, retreated ice sheets and sea levels comparable to those of today.

Palaeogenomics

The genome-wide analysis of DNA from ancient organisms.

past demographic history influences the degree of mutational load in threatened populations. In the Sumatran rhinoceros (*Dicerorhinus sumatrensis*) and Grauer's gorilla (*Gorilla beringei graueri*), marked increases in mutational load could be inferred from aDNA from museum specimens following genetic bottlenecks within the past few decades^{144,145}. However, in several other species, such as Alpine ibex (*Capra ibex*), Iberian lynx (*Lynx pardinus*) and kākāpō (*Strigops habroptila*), mutational load was instead found to be lower in contemporary populations compared to past populations^{151–155}. A closer look at their genetic past revealed that these populations had experienced low genetic diversity for a long time, allowing for purging of deleterious mutations¹⁵⁶. Therefore, demographic history can be shown to influence the extinction risk of taxa, which indicates that populations can avoid accumulation of deleterious mutations if bottlenecks are gradual or followed by rapid population recovery. Such knowledge is important for inferring an accurate conservation status of species and guiding conservation actions.

aDNA can also be used to assess and monitor the success of modern-day management programmes. For example, temporal genomic data from the crested ibis (*Nipponia nippon*) show that although the population has increased in size owing to recent conservation efforts, the species suffers from a severe loss of genetic diversity and the population needs to be continuously monitored to ensure its survival¹⁵⁷.

Summary and future directions

Rapid advances in palaeogenomic techniques are making the acquisition of information about diversity possible across long time periods, from deep time to the present, and at all hierarchical levels of

biodiversity. The unique characteristics of palaeogenomic data allows researchers to generate a wealth of information on past ecological crises and extinctions, as well as on past conditions of the resilience and stability of ecosystems, species assemblages and populations. The possibility of tracking the history of species and ecosystems back through the past and of identifying periods of stability and change, and their causes, offers huge potential for integration into current conservation efforts. Importantly, the unique nature of palaeogenomic data combines both ecological and evolutionary information at different levels of spatial and temporal resolution, which could enhance our understanding of past biodiversity changes and help us to develop forecasts of biodiversity under global change, at the ecosystem, species and genetic levels.

Palaeogenomic data can be incorporated into ecological niche models¹⁵⁸ and combined with data on changes in intraspecific genetic diversity. Together, these models could inform correlations of population size and climate¹⁵⁹, providing complementary evidence for future ecological trajectories. Causal relationships inferred from joint time series data of organisms based on palaeogenomics and abiotic factors could furthermore be experimentally validated for certain timepoints to identify the parameters and processes to be used in projections of future biodiversity change¹⁶⁰. Palaeogenomic data could also provide ground truths for projections on the significance of future diversity change for ecosystem stability. For example, if a model predicts that the relationship of diversity and stability of ecosystems depends on the form of population growth¹⁶¹, this could be investigated using time series data.

In this Review, we have discussed how the use of palaeogenomic data can be instrumental in elucidating which drivers of biodiversity change have been the most important across different time periods, and can pinpoint when and where direct anthropogenic impact has been decisive. For example, aDNA data combined with palaeoclimate variables and human occurrence data has been incorporated into a model to investigate the factors driving distributions of mammals over the past 50,000 years across the Arctic⁸². The model indicated that, in this case, human occurrence alone had little effect on the presence of herbivores, whereas precipitation, a climate variable, was the best predictor for their distribution. In other cases, palaeogenomic data identifies human land use as the decisive factor for structural changes in terrestrial¹⁸⁹ and aquatic¹⁶² biodiversity. A comparative palaeogenomic investigation of biodiversity trajectories from times predating human impact (such as previous interglacial periods) and from the current interglacial period (the Holocene) could more fully evaluate the long-term role of humans in shaping biodiversity. This approach could also reveal which taxa or phylogenetic groups are affected most by human pressure and environmental changes, and which show high resilience throughout their history, thereby indicating future vulnerabilities.

Finally, we have discussed how palaeogenomics can not only deliver an understanding of crises and extinctions, but also of the processes that counteract these, such as purging of deleterious mutations and identifying the times and causes of increasing biodiversity in the past. By combining data on environmental changes from geology and archaeology with the full range of possibilities for palaeogenomic data acquisition across species and ecosystems, we can contribute substantially to developing avenues for conservation.

Palaeogenomics is a valuable tool in biodiversity research and will increasingly become more useful as our capacity to generate more and higher-quality aDNA from different periods and sources grows.

At present, palaeogenomic studies from the industrial era are focused mostly on species and genetic biodiversity, while studies of ecosystems are lacking, even though current techniques are capable of capturing ecosystem biodiversity from much further back in time. Deep-time studies are sparse as well, especially studies focusing on loss of genetic biodiversity, but these will probably increase in number as aDNA techniques improve. The increasing availability of palaeogenomic as well as reference genomic data and their integration into accessible databases will allow large-scale investigations of past biodiversity changes across taxa, time and space. Palaeogenomics already contributes decisively to our knowledge on past biodiversity change and it will be a crucial tool in future biodiversity research.

Published online: 18 March 2025

References

- Dalén, L., Heintzman, P. D., Kapp, J. D. & Shapiro, B. Deep-time paleogenomics and the limits of DNA survival. *Science* **382**, 48–53 (2023).
- van der Valk, T. et al. Million-year-old DNA sheds light on the genomic history of mammoths. *Nature* **591**, 265–269 (2021).
- Kjær, K. H. et al. A 2-million-year-old ecosystem in Greenland uncovered by environmental DNA. *Nature* **612**, 283–291 (2022).
- Green, E. J. & Speller, C. F. Novel substrates as sources of ancient DNA: prospects and hurdles. *Genes* **8**, 180 (2017).
- Epp, L. S., Zimmermann, H. H. & Stooft-Leichsenring, K. R. Sampling and extraction of ancient DNA from sediments. *Methods Mol. Biol.* **1963**, 31–44 (2019).
- Wagner, S. et al. High-throughput DNA sequencing of ancient wood. *Mol. Ecol.* **27**, 1138–1154 (2018).
- Ceballos, G. et al. Accelerated modern human-induced species losses: entering the sixth mass extinction. *Sci. Adv.* **1**, e1400253 (2015).
- Ceballos, G., Ehrlich, P. R. & Dirzo, R. Biological annihilation via the ongoing sixth mass extinction signaled by vertebrate population losses and declines. *Proc. Natl Acad. Sci. USA* **114**, E6089–E6096 (2017).
- Dasgupta, P. S. & Ehrlich, P. R. In *Biological Extinction: New Perspectives* (eds Dasgupta, P., Raven, P. & McIvor, A.) 262–284 (Cambridge Univ. Press, 2019).
- Cowie, R. H., Bouchet, P. & Fontaine, B. The sixth mass extinction: fact, fiction or speculation? *Biol. Rev. Camb. Phil. Soc.* **97**, 640–663 (2022).
- Clarke, C. L. et al. Steppe–tundra composition and deglacial floristic turnover in interior Alaska revealed by sedimentary ancient DNA (sedaDNA). *Quat. Sci. Rev.* **334**, 108672 (2024).
- Voldstad, L. H. et al. A complete Holocene lake sediment ancient DNA record reveals long-standing high Arctic plant diversity hotspot in northern Svalbard. *Quat. Sci. Rev.* **234**, 106207 (2020).
- Dehasque, M. et al. Temporal dynamics of woolly mammoth genome erosion prior to extinction. *Cell* **187**, 3531–3540.e13 (2024).
- Diez-Del-Molino, D. et al. Genomics of adaptive evolution in the woolly mammoth. *Curr. Biol.* **33**, 1753–1764.e4 (2023).
- McGhee, G. R., Sheehan, P. M., Bottjer, D. J. & Droser, M. L. Ecological ranking of Phanerozoic biodiversity crises: ecological and taxonomic severities are decoupled. *Palaeogeogr. Palaeoclimatol. Palaeoecol.* **211**, 289–297 (2004).
- Barnosky, A. D. et al. Has the Earth's sixth mass extinction already arrived? *Nature* **471**, 51–57 (2011).
- Stewart, J. R., Lister, A. M., Barnes, I. & Dalén, L. Refugia revisited: individualistic responses of species in space and time. *Proc. Biol. Sci.* **277**, 661–671 (2010).
- Barnosky, A. D., Koch, P. L., Feranec, R. S., Wing, S. L. & Shabel, A. B. Assessing the causes of late Pleistocene extinctions on the continents. *Science* **306**, 70–75 (2004).
- Svenning, J.-C. et al. The late-Quaternary megafauna extinctions: patterns, causes, ecological consequences and implications for ecosystem management in the Anthropocene. *Camb. Prisms Extinct.* **2**, e5 (2024).
- Tóth, A. B. et al. Reorganization of surviving mammal communities after the end-Pleistocene megafaunal extinction. *Science* **365**, 1305–1308 (2019).
- Galetti, M. et al. Ecological and evolutionary legacy of megafauna extinctions. *Biol. Rev. Camb. Phil. Soc.* **93**, 845–862 (2018).
- IPBES. *Summary for Policymakers of the Global Assessment Report on Biodiversity and Ecosystem Services of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services* (IPBES secretariat, 2019).
- Exposito-Alonso, M. et al. Genetic diversity loss in the Anthropocene. *Science* **377**, 1431–1435 (2022).
- Willi, Y., Van Buskirk, J. & Hoffmann, A. A. Limits to the adaptive potential of small populations. *Annu. Rev. Ecol. Syst.* **37**, 433–458 (2006).
- Higuchi, R., Bowman, B., Freiberger, M., Ryder, O. A. & Wilson, A. C. DNA sequences from the quagga, an extinct member of the horse family. *Nature* **312**, 282–284 (1984).
- Cooper, R. A. et al. Completeness of the fossil record: estimating losses due to small body size. *Geology* **34**, 241 (2006).
- Smith, A. B. Large-scale heterogeneity of the fossil record: implications for Phanerozoic biodiversity studies. *Phil. Trans. R. Soc. Lond. B* **356**, 351–367 (2001).
- Thomas, R. H., Schaffner, W., Wilson, A. C. & Pääbo, S. DNA phylogeny of the extinct marsupial wolf. *Nature* **340**, 465–467 (1989).
- Cooper, A. et al. Independent origins of New Zealand moas and kiwis. *Proc. Natl Acad. Sci. USA* **89**, 8741–8744 (1992).
- Wayne, R. K., Leonard, J. A. & Cooper, A. Full of sound and fury: history of ancient DNA. *Annu. Rev. Ecol. Syst.* **30**, 457–477 (1999).
- Knapp, M. & Hofreiter, M. Next generation sequencing of ancient DNA: requirements, strategies and perspectives. *Genes* **1**, 227–243 (2010).
- Dalal, V., Pasupuleti, N., Chaubey, G., Rai, N. & Shinde, V. Advancements and challenges in ancient DNA research: bridging the global North–South divide. *Genes* **14**, 479 (2023).
- Breed, M. F. et al. The potential of genomics for restoring ecosystems and biodiversity. *Nat. Rev. Genet.* **20**, 615–628 (2019).
- Theissinger, K. et al. How genomics can help biodiversity conservation. *Trends Genet.* **39**, 545–559 (2023).
- Hogg, C. J. Translating genomic advances into biodiversity conservation. *Nat. Rev. Genet.* **25**, 362–373 (2024).
- Sawyer, S., Krause, J., Guschanski, K., Savolainen, V. & Pääbo, S. Temporal patterns of nucleotide misincorporations and DNA fragmentation in ancient DNA. *PLoS ONE* **7**, e34131 (2012).
- Kistler, L., Ware, R., Smith, O., Collins, M. & Allaby, R. G. A new model for ancient DNA decay based on paleogenomic meta-analysis. *Nucleic Acids Res.* **45**, 6310–6320 (2017).
- Mullin, V. E. et al. First large-scale quantification study of DNA preservation in insects from natural history collections using genome-wide sequencing. *Methods Ecol. Evol.* **14**, 360–371 (2023).
- Llomas, B. et al. From the field to the laboratory: controlling DNA contamination in human ancient DNA research in the high-throughput sequencing era. *STAR Sci. Technol. Archaeol. Res.* **3**, 1–14 (2017).
- Dabney, J., Meyer, M. & Pääbo, S. Ancient DNA damage. *Cold Spring Harb. Persp. Biol.* **5**, a012567 (2013).
- Rohland, N., Glocke, I., Aximu-Petri, A. & Meyer, M. Extraction of highly degraded DNA from ancient bones, teeth and sediments for high-throughput sequencing. *Nat. Protoc.* **13**, 2447–2461 (2018).
- Epp, L. S., Stooft, K. R., Trauth, M. H. & Tiedemann, R. Historical genetics on a sediment core from a Kenyan lake: intraspecific genotype turnover in a tropical rotifer is related to past environmental changes. *J. Paleolimnol.* **43**, 939–954 (2010).
- Stooft-Leichsenring, K. R., Epp, L. S., Trauth, M. H. & Tiedemann, R. Hidden diversity in diatoms of Kenyan Lake Naivasha: a genetic approach detects temporal variation. *Mol. Ecol.* **21**, 1918–1930 (2012).
- Dommain, R. et al. The challenges of reconstructing tropical biodiversity with sedimentary ancient DNA: a 2200-year-long metagenomic record from Bwindi Impenetrable Forest, Uganda. *Front. Ecol. Evol.* **8**, 218 (2020).
- Epp, L. S. A global perspective for biodiversity history with ancient environmental DNA. *Mol. Ecol.* **28**, 2456–2458 (2019).
- Pedersen, M. W. et al. Ancient and modern environmental DNA. *Phil. Trans. R. Soc. Lond. B* **370**, 20130383 (2015).
- Schlumbaum, A., Tensen, M. & Jaenicke-Després, V. Ancient plant DNA in archaeobotany. *Veg. Hist. Archaeobot.* **17**, 233–244 (2008).
- Murchie, T. J. et al. Pleistocene mitogenomes reconstructed from the environmental DNA of permafrost sediments. *Curr. Biol.* **32**, 851–860.e7 (2022).
- Schulte, L. et al. Larix species range dynamics in Siberia since the Last Glacial captured from sedimentary ancient DNA. *Commun. Biol.* **5**, 570 (2022).
- Heintzman, P. D. et al. In *Tracking Environmental Change Using Lake Sediments* (eds Capo, E., Barouillet, C. & Smol, J. P.) 53–84 (Springer International, 2023).
- Pedersen, M. W. et al. Postglacial viability and colonization in North America's ice-free corridor. *Nature* **537**, 45–49 (2016).
- Willerslev, E. et al. Fifty thousand years of Arctic vegetation and megafaunal diet. *Nature* **506**, 47–51 (2014).
- Taberlet, P., Coissac, E., Hajibabaei, M. & Rieseberg, L. H. Environmental DNA: environmental DNA. *Mol. Ecol.* **21**, 1789–1793 (2012).
- Mamanova, L. et al. Target-enrichment strategies for next-generation sequencing. *Nat. Methods* **7**, 111–118 (2010).
- Murchie, T. J. et al. Collapse of the mammoth-steppe in central Yukon as revealed by ancient environmental DNA. *Nat. Commun.* **12**, 7120 (2021).
- Schulte, L. et al. Hybridization capture of larch (*Larix* Mill.) chloroplast genomes from sedimentary ancient DNA reveals past changes of Siberian forest. *Mol. Ecol. Resour.* **21**, 801–815 (2021).
- Cribdon, B., Ware, R., Smith, O., Gaffney, V. & Allaby, R. G. PIA: more accurate taxonomic assignment of metagenomic data demonstrated on sedaDNA from the North Sea. *Front. Ecol. Evol.* **8**, 506594 (2020).
- González Fortes, G. & Pajmans, J. L. A. Whole-genome capture of ancient DNA using homemade baits. *Methods Mol. Biol.* **1963**, 93–105 (2019).
- Soares, A. E. R. Hybridization capture of ancient DNA using RNA baits. *Methods Mol. Biol.* **1963**, 121–128 (2019).
- Poinar, H. N. et al. Metagenomics to paleogenomics: large-scale sequencing of mammoth DNA. *Science* **311**, 392–394 (2006).
- Carpenter, M. L. et al. Pulling out the 1%: whole-genome capture for the targeted enrichment of ancient DNA sequencing libraries. *Am. J. Hum. Genet.* **93**, 852–864 (2013).

62. Nistelberger, H. M., Smith, O., Wales, N., Star, B. & Boessenkool, S. The efficacy of high-throughput sequencing and target enrichment on charred archaeobotanical remains. *Sci. Rep.* **6**, 37347 (2016).
63. Bunce, M. et al. Extreme reversed sexual size dimorphism in the extinct New Zealand moa *Dinornis*. *Nature* **425**, 172–175 (2003).
64. Huynen, L., Millar, C. D., Scofield, R. P. & Lambert, D. M. Nuclear DNA sequences detect species limits in ancient moa. *Nature* **425**, 175–178 (2003).
65. da Fonseca, R. R. et al. The origin and evolution of maize in the Southwestern United States. *Nat. Plants* **1**, 14003 (2015).
66. Park, S. D. E. et al. Genome sequencing of the extinct Eurasian wild aurochs, *Bos primigenius*, illuminates the phylogeography and evolution of cattle. *Genome Biol.* **16**, 234 (2015).
67. Wang, M.-S. et al. A polar bear paleogenome reveals extensive ancient gene flow from polar bears into brown bears. *Nat. Ecol. Evol.* **6**, 936–944 (2022).
68. Casas-Marce, M. et al. Spatiotemporal dynamics of genetic variation in the Iberian lynx along its path to extinction reconstructed with ancient DNA. *Mol. Biol. Evol.* **34**, 2893–2907 (2017).
69. Kellner, F. L. et al. A palaeogenomic investigation of overharvest implications in an endemic wild reindeer subspecies. *Mol. Ecol.* **33**, e12724 (2024).
70. Palkopoulou, E. et al. Synchronous genetic turnovers across Western Eurasia in Late Pleistocene collared lemmings. *Glob. Chang. Biol.* **22**, 1710–1721 (2016).
71. Baca, M. et al. Ancient DNA reveals interstadials as a driver of common vole population dynamics during the last glacial period. *J. Biogeogr.* **50**, 183–196 (2023).
72. Parducci, L., Suyama, Y., Lascoux, M. & Bennett, K. D. Ancient DNA from pollen: a genetic record of population history in Scots pine. *Mol. Ecol.* **14**, 2873–2882 (2005).
73. Ehrlich, P. R., Kremen, C. & Ehrlich, A. H. In *Encyclopedia of Biodiversity* 2nd edn (ed. Levin, S. A.) 153–161 (Elsevier, 2013).
74. Berends, C. J., Köhler, P., Lourens, L. J. & van de Wal, R. S. W. On the cause of the mid-Pleistocene transition. *Rev. Geophys.* **59**, e2020RG000727 (2021).
75. Willerslev, E. et al. Ancient biomolecules from deep ice cores reveal a forested southern Greenland. *Science* **317**, 111–114 (2007).
76. Crump, S. E. et al. Ancient plant DNA reveals High Arctic greening during the Last Interglacial. *Proc. Natl Acad. Sci. USA* **118**, e2019069118 (2021).
77. Courtin, J. et al. Pleistocene glacial and interglacial ecosystems inferred from ancient DNA analyses of permafrost sediments from Batagay megaslump, East Siberia. *Environ. DNA* **4**, 1265–1283 (2022).
78. Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Summary for Policymakers. *IPCC* https://www.ipcc.ch/report/ar6/syr/downloads/report/IPCC_AR6_SYR_SPM.pdf (2023).
79. Park, T. et al. Changes in growing season duration and productivity of northern vegetation inferred from long-term remote sensing data. *Environ. Res. Lett.* **11**, 084001 (2016).
80. Alsos, I. G. et al. Plant DNA metabarcoding of lake sediments: how does it represent the contemporary vegetation. *PLoS ONE* **13**, e0195403 (2018).
81. Stivirs, N. et al. Biotic turnover rates during the Pleistocene–Holocene transition. *Quat. Sci. Rev.* **151**, 100–110 (2016).
82. Wang, Y. et al. Late Quaternary dynamics of Arctic biota from ancient environmental genomics. *Nature* **600**, 86–92 (2021).
83. Jeffers, E. S. et al. Plant controls on Late Quaternary whole ecosystem structure and function. *Ecol. Lett.* **21**, 814–825 (2018).
84. Seersholm, F. V. et al. Rapid range shifts and megafaunal extinctions associated with late Pleistocene climate change. *Nat. Commun.* **11**, 2770 (2020).
85. Rijal, D. P. et al. Sedimentary ancient DNA shows terrestrial plant richness continuously increased over the Holocene in northern Fennoscandia. *Sci. Adv.* **7**, eabf9557 (2021).
86. Alsos, I. G. et al. Postglacial species arrival and diversity buildup of northern ecosystems took millennia. *Sci. Adv.* **8**, eabo7434 (2022).
87. Boilard, A. et al. Ancient DNA and osteological analyses of a unique paleo-archive reveal Early Holocene faunal expansion into the Scandinavian Arctic. *Sci. Adv.* **10**, eadk3032 (2024).
88. Clarke, C. L. et al. Persistence of Arctic–alpine flora during 24,000 years of environmental change in the polar Urals. *Sci. Rep.* **9**, 19613 (2019).
89. Garcés-Pastor, S. et al. High resolution ancient sedimentary DNA shows that alpine plant diversity is associated with human land use and climate change. *Nat. Commun.* **13**, 6559 (2022).
90. Pansu, J. et al. Reconstructing long-term human impacts on plant communities: an ecological approach based on lake sediment DNA. *Mol. Ecol.* **24**, 1485–1498 (2015).
91. ter Schure, A. T. M. et al. Anthropogenic and environmental drivers of vegetation change in southeastern Norway during the Holocene. *Quat. Sci. Rev.* **270**, 107175 (2021).
92. Lindahl, T. Instability and decay of the primary structure of DNA. *Nature* **362**, 709–715 (1993).
93. Boessenkool, S. et al. Use of ancient sedimentary DNA as a novel conservation tool for high-altitude tropical biodiversity. *Conserv. Biol.* **28**, 446–455 (2014).
94. Jia, W. et al. Preservation of sedimentary plant DNA is related to lake water chemistry. *Environ. DNA* **4**, 425–439 (2022).
95. Isbell, F. et al. Expert perspectives on global biodiversity loss and its drivers and impacts on people. *Front. Ecol. Environ.* **21**, 94–103 (2023).
96. Ellis, E. C., Klein Goldewijk, K., Siebert, S., Lightman, D. & Ramankutty, N. Anthropogenic transformation of the biomes, 1700 to 2000. *Glob. Ecol. Biogeogr.* **19**, 589–606 (2010).
97. Ruddiman, W. F. The Anthropocene. *Annu. Rev. Earth Planet. Sci.* **41**, 45–68 (2013).
98. Johnson, C. N. et al. Biodiversity losses and conservation responses in the Anthropocene. *Science* **356**, 270–275 (2017).
99. Turvey, S. T. & Cress, J. J. Extinction in the Anthropocene. *Curr. Biol.* **29**, R982–R986 (2019).
100. Capo, E. et al. Tracking a century of changes in microbial eukaryotic diversity in lakes driven by nutrient enrichment and climate warming. *Environ. Microbiol.* **19**, 2873–2892 (2017).
101. Li, F., Zhang, X., Xie, Y. & Wang, J. Sedimentary DNA reveals over 150 years of ecosystem change by human activities in Lake Chao, China. *Environ. Int.* **133**, 105214 (2019).
102. Ibrahim, A. et al. Anthropogenic impact on the historical phytoplankton community of lake conformance reconstructed by multimarker analysis of sediment-core environmental DNA. *Mol. Ecol.* **30**, 3040–3056 (2021).
103. Bauer, S. & Høye, B. J. Migratory animals couple biodiversity and ecosystem functioning worldwide. *Science* **344**, 1242552 (2014).
104. Tucker, M. A. et al. Moving in the Anthropocene: global reductions in terrestrial mammalian movements. *Science* **359**, 466–469 (2018).
105. Barouillet, C. et al. Investigating the effects of anthropogenic stressors on lake biota using sedimentary DNA. *Freshw. Biol.* **68**, 1799–1817 (2022).
106. Monchamp, M.-E. et al. Homogenization of lake cyanobacterial communities over a century of climate change and eutrophication. *Nat. Ecol. Evol.* **2**, 317–324 (2018).
107. Barouillet, C. et al. Paleoreconstructions of ciliate communities reveal long-term ecological changes in temperate lakes. *Sci. Rep.* **12**, 7899 (2022).
108. Armbricht, L. et al. Ancient marine sediment DNA reveals diatom transition in Antarctica. *Nat. Commun.* **13**, 5787 (2022).
109. Bagousse-Pinguet, Y. L. et al. Phylogenetic, functional, and taxonomic richness have both positive and negative effects on ecosystem multifunctionality. *Proc. Natl Acad. Sci. USA* **116**, 8419–8424 (2019).
110. Tilman, D., Isbell, F. & Cowles, J. M. Biodiversity and ecosystem functioning. *Annu. Rev. Ecol. Syst.* **45**, 471–493 (2014).
111. Isbell, F. et al. Biodiversity increases the resistance of ecosystem productivity to climate extremes. *Nature* **526**, 574–577 (2015).
112. B-Béres, V. et al. Ecosystem services provided by freshwater and marine diatoms. *Hydrobiologia* **850**, 2707–2733 (2023).
113. Murray, D. C. et al. Scrapheap challenge: a novel bulk-bone metabarcoding method to investigate ancient DNA in faunal assemblages. *Sci. Rep.* **3**, 3371 (2013).
114. Woods, R., Barnes, I., Brace, S. & Turvey, S. T. Ancient DNA suggests single colonization and within-archipelago diversification of Caribbean caviomorph rodents. *Mol. Biol. Evol.* **38**, 84–95 (2021).
115. Dalén, L. et al. Ancient DNA reveals lack of postglacial habitat tracking in the arctic fox. *Proc. Natl Acad. Sci. USA* **104**, 6726–6729 (2007).
116. Foote, A. D. et al. Ancient DNA reveals that bowhead whale lineages survived Late Pleistocene climate change and habitat shifts. *Nat. Commun.* **4**, 1677 (2013).
117. Meiri, M., Lister, A., Kosintsev, P., Zazula, G. & Barnes, I. Population dynamics and range shifts of moose (*Alces alces*) during the Late Quaternary. *J. Biogeogr.* **47**, 2223–2234 (2020).
118. Wang, Y. et al. Author Correction: Late Quaternary dynamics of Arctic biota from ancient environmental genomics. *Nature* **610**, E5 (2022).
119. Haile, J. et al. Ancient DNA reveals late survival of mammoth and horse in interior Alaska. *Proc. Natl Acad. Sci. USA* **106**, 22352–22357 (2009).
120. Graham, R. W. et al. Timing and causes of mid-Holocene mammoth extinction on St. Paul Island, Alaska. *Proc. Natl Acad. Sci. USA* **113**, 9310–9314 (2016).
121. Jaureguiberry, P. et al. The direct drivers of recent global anthropogenic biodiversity loss. *Sci. Adv.* **8**, eabm9982 (2022).
122. White, L. C., Mitchell, K. J. & Austin, J. J. Ancient mitochondrial genomes reveal the demographic history and phylogeography of the extinct, enigmatic thylacine (*Thylacinus cynocephalus*). *J. Biogeogr.* **45**, 1–13 (2018).
123. Murray, G. G. R. et al. Natural selection shaped the rise and fall of passenger pigeon genomic diversity. *Science* **358**, 951–954 (2017).
124. Thomas, J. E. et al. Demographic reconstruction from ancient DNA supports rapid extinction of the great auk. *eLife* **8**, e47509 (2019).
125. Sharko, F. S. et al. Steller's sea cow genome suggests this species began going extinct before the arrival of Paleolithic humans. *Nat. Commun.* **12**, 2215 (2021).
126. Yan, D. et al. Sedimentary DNA reveals phytoplankton diversity loss in a deep maar lake during the Anthropocene. *Limnol. Oceanogr.* **69**, 1299–1315 (2024).
127. Schmidt, A. et al. Decoding the Baltic Sea's past and present: a simple molecular index for ecosystem assessment. *Ecol. Indic.* **166**, 112494 (2024).
128. Siano, R. et al. Sediment archives reveal irreversible shifts in plankton communities after World War II and agricultural pollution. *Curr. Biol.* **31**, 2682–2689.e7 (2021).
129. Shaw, J. L. A., Weyrich, L. S., Hallegraeff, G. & Cooper, A. Retrospective eDNA assessment of potentially harmful algae in historical ship ballast tank and marine port sediments. *Mol. Ecol.* **28**, 2476–2485 (2019).
130. Ficetola, G. F. et al. DNA from lake sediments reveals long-term ecosystem changes after a biological invasion. *Sci. Adv.* **4**, eaar4292 (2018).
131. Daszak, P., Cunningham, A. A. & Hyatt, A. D. Emerging infectious diseases of wildlife — threats to biodiversity and human health. *Science* **287**, 443–449 (2000).
132. Kelly, L. T. et al. Fire and biodiversity in the Anthropocene. *Science* **370**, eabb0355 (2020).
133. Sims, D., Sudbery, I., Illott, N. E., Heger, A. & Ponting, C. P. Sequencing depth and coverage: key considerations in genomic analyses. *Nat. Rev. Genet.* **15**, 121–132 (2014).
134. Depaulis, F., Orlando, L. & Hänni, C. Using classical population genetics tools with heterochronous data: time matters! *PLoS ONE* **4**, e5541 (2009).

135. Orlando, L. et al. Recalibrating *Equus* evolution using the genome sequence of an early Middle Pleistocene horse. *Nature* **499**, 74–78 (2013).
136. Barlow, A. et al. Middle Pleistocene genome calibrates a revised evolutionary history of extinct cave bears. *Curr. Biol.* **31**, 1771–1779.e7 (2021).
137. Lord, E. et al. Population dynamics and demographic history of Eurasian collared lemmings. *BMC Ecol. Evol.* **22**, 126 (2022).
138. Baca, M. et al. Ancient DNA of narrow-headed vole reveal common features of the Late Pleistocene population dynamics in cold-adapted small mammals. *Proc. Biol. Sci.* **290**, 20222238 (2023).
139. Loog, L. et al. Ancient DNA suggests modern wolves trace their origin to a Late Pleistocene expansion from Beringia. *Mol. Ecol.* **29**, 1596–1610 (2020).
140. Cahill, J. A. et al. Genomic evidence of widespread admixture from polar bears into brown bears during the last ice age. *Mol. Biol. Evol.* **35**, 1120–1129 (2018).
141. Marr, M. M., Brace, S., Schreve, D. C. & Barnes, I. Identifying source populations for the reintroduction of the Eurasian beaver, *Castor fiber* L. 1758, into Britain: evidence from ancient DNA. *Sci. Rep.* **8**, 2708 (2018).
142. Palkopoulou, E. et al. Complete genomes reveal signatures of demographic and genetic declines in the woolly mammoth. *Curr. Biol.* **25**, 1395–1400 (2015).
143. Lord, E. et al. Pre-extinction demographic stability and genomic signatures of adaptation in the woolly rhinoceros. *Curr. Biol.* **30**, 3871–3879.e7 (2020).
144. von Seth, J. et al. Genomic insights into the conservation status of the world's last remaining Sumatran rhinoceros populations. *Nat. Commun.* **12**, 2393 (2021).
145. van der Valk, T., Díez-Del-Molino, D., Marques-Bonet, T., Guschanski, K. & Dalén, L. Historical genomes reveal the genomic consequences of recent population decline in Eastern Gorillas. *Curr. Biol.* **29**, 165–170.e6 (2019).
146. Díez-del-Molino, D., Sánchez-Barreiro, F., Barnes, I., Gilbert, M. T. P. & Dalén, L. Quantifying temporal genomic erosion in endangered species. *Trends Ecol. Evol.* **33**, 176–185 (2018).
147. Duncan, R. P., Boyer, A. G. & Blackburn, T. M. Magnitude and variation of prehistoric bird extinctions in the Pacific. *Proc. Natl Acad. Sci. USA* **110**, 6436–6441 (2013).
148. Wood, J. R. et al. Island extinctions: processes, patterns, and potential for ecosystem restoration. *Environ. Conserv.* **44**, 348–358 (2017).
149. Fernández-Palacios, J. M. et al. Scientists' warning — the outstanding biodiversity of islands is in peril. *Glob. Ecol. Conserv.* **31**, e01847 (2021).
150. Roycroft, E. et al. Museum genomics reveals the rapid decline and extinction of Australian rodents since European settlement. *Proc. Natl Acad. Sci. USA* **118**, e2021390118 (2021).
151. Mathur, S. & DeWoody, J. A. Genetic load has potential in large populations but is realized in small inbred populations. *Evol. Appl.* **14**, 1540–1557 (2021).
152. Kleinman-Ruiz, D. et al. Purging of deleterious burden in the endangered Iberian lynx. *Proc. Natl Acad. Sci. USA* **119**, e2110614119 (2022).
153. Grossen, C., Guillaume, F., Keller, L. F. & Croll, D. Purging of highly deleterious mutations through severe bottlenecks in Alpine ibex. *Nat. Commun.* **11**, 1001 (2020).
154. Xue, Y. et al. Mountain gorilla genomes reveal the impact of long-term population decline and inbreeding. *Science* **348**, 242–245 (2015).
155. Dussex, N. et al. Population genomics of the critically endangered kākāpō. *Cell Genom.* **1**, 100002 (2021).
156. Dussex, N., Morales, H. E., Grossen, C., Dalén, L. & van Oosterhout, C. Purging and accumulation of genetic load in conservation. *Trends Ecol. Evol.* **38**, 961–969 (2023).
157. Feng, S. et al. The genomic footprints of the fall and recovery of the crested ibis. *Curr. Biol.* **29**, 340–349.e7 (2019).
158. Fordham, D. A., Brook, B. W., Moritz, C. & Nogués-Bravo, D. Better forecasts of range dynamics using genetic data. *Trends Ecol. Evol.* **29**, 436–443 (2014).
159. Prost, S. et al. Losing ground: past history and future fate of Arctic small mammals in a changing climate. *Glob. Change Biol.* **19**, 1854–1864 (2013).
160. Eastwood, N. et al. The Time Machine framework: monitoring and prediction of biodiversity loss. *Trends Ecol. Evol.* **37**, 138–146 (2022).
161. Hatton, I. A., Mazzarisi, O., Altieri, A. & Smerlak, M. Diversity begets stability: sublinear growth and competitive coexistence across ecosystems. *Science* **383**, eadg8488 (2024).
162. Nwosu, E. C. et al. Early human impact on lake cyanobacteria revealed by a Holocene record of sedimentary ancient DNA. *Commun. Biol.* **6**, 72 (2023).
163. Jouzel, J. et al. Orbital and millennial Antarctic climate variability over the past 800,000 years. *Science* **317**, 793–796 (2007).
164. Andersen, K. K. et al. High-resolution record of Northern Hemisphere climate extending into the last interglacial period. *Nature* **431**, 147–151 (2004).
165. Morice, C. P., Kennedy, J. J., Rayner, N. A. & Jones, P. D. Quantifying uncertainties in global and regional temperature change using an ensemble of observational estimates: the HadCRUT4 data set. *J. Geophys. Res. Atmos.* <https://doi.org/10.1029/2011JD017187> (2012).
166. Lisiecki, L. E. & Raymo, M. E. A. Pliocene–Pleistocene stack of 57 globally distributed benthic $\delta^{18}O$ records. *Paleoceanography* <https://doi.org/10.1029/2004PA001071> (2005).
167. IPCC. *Climate Change 2013: The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change* (eds Stocker, T. F. et al.) (Cambridge Univ. Press, 2013).
168. Burke, K. D. et al. Pliocene and Eocene provide best analogs for near-future climates. *Proc. Natl Acad. Sci. USA* **115**, 13288–13293 (2018).
169. Ter Schure, A. T. M. et al. Sedimentary ancient DNA metabarcoding as a tool for assessing prehistoric plant use at the Upper Paleolithic cave site Aghitu-3, Armenia. *J. Hum. Evol.* **172**, 103258 (2022).
170. Parducci, L. et al. Ancient plant DNA in lake sediments. *N. Phytol.* **214**, 924–942 (2017).

Acknowledgements

S. Boessenkool acknowledges support from the Research Council of Norway (grant 314464). The authors thank S. Nylin and K. Norén, as well as the students in course BL7075 at Stockholm University for providing feedback on earlier versions of the manuscript.

Author contributions

L.D., A.L., D.D.d.M. and P.D.H. conceived the review and A.L. wrote the first draft. All authors contributed to discussion of the content, conceptualized the figures, substantially reviewed and edited the manuscript before submission, and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Peer review information *Nature Reviews Biodiversity* thanks Tyler Murchie and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Related links

Convention on Biological Diversity: <http://www.cbd.int/>
Earth BioGenome Project: <https://www.earthbiogenome.org/>
The Intergovernmental Panel on Climate Change: <https://www.ipcc.ch/>

© Springer Nature Limited 2025